DSC5001: Statistical Machine Learning I

Midterm Examination – Solutions

Semester A, 2026/27

Total: 100 marks

Part I Answer Key

Question	1	2	3	4	5	6	7	8
Answer	B	C	B	C	B	C	B	C
Question	9	10	11	12	13	14	15	16
Answer	A	B	C	A	B	C	A	C

Part	Marks
Part I: Multiple Choice	80
Question 17	10
Question 18	10
Total	100

Part I: Multiple-Choice Solutions

Question 1. Answer: B

By Bayes' rule,

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)}.$$

Hence

$$P(D \mid +) = \frac{0.90(0.05)}{0.90(0.05) + 0.10(0.95)} = \frac{0.045}{0.14} \approx \boxed{0.321}.$$

Question 2. Answer: C

Using

$$\mathrm{Var}(aX + bY) = a^2 \,\mathrm{Var}(X) + b^2 \,\mathrm{Var}(Y) + 2ab \,\mathrm{Cov}(X, Y),$$

with $a = 2$ and $b = -1$,

$$\mathrm{Var}(2X - Y) = 4(4) + 9 + 2(2)(-1)(-1) = 16 + 9 + 4 = \boxed{29}.$$

Question 3. Answer: B

If X and Y are independent and have finite second moments,

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y],$$

so

$$\mathrm{Cov}(X, Y) = 0.$$

The converse is not true in general.

Question 4. Answer: C

For a Bernoulli sample with h successes in n trials,

$$\ell(p) = h \log p + (n - h) \log(1 - p),$$

so

$$\ell''(p) = -\frac{h}{p^2} - \frac{n - h}{(1 - p)^2}.$$

For D_A, $h = 3$, $n = 4$, and $p = 0.75$:

$$|\ell''_A(0.75)| = \frac{3}{0.75^2} + \frac{1}{0.25^2} = \frac{64}{3}.$$

For D_B, $h = 30$, $n = 40$:

$$|\ell''_B(0.75)| = \frac{30}{0.75^2} + \frac{10}{0.25^2} = \frac{640}{3}.$$

Therefore,

$$\frac{|\ell''_B(0.75)|}{|\ell''_A(0.75)|} = \boxed{10}.$$

Question 5. Answer: B

The MLE of the Gaussian mean is

$$\widehat{\mu} = \bar{x} = \frac{-1 + 1 + 3 + 5}{4} = 2.$$

The MLE of the variance uses denominator n:

$$\widehat{\sigma}^2 = \frac{1}{4}\left[(-1 - 2)^2 + (1 - 2)^2 + (3 - 2)^2 + (5 - 2)^2\right].$$

Thus

$$\widehat{\sigma}^2 = \frac{9+1+1+9}{4} = \boxed{5}.$$

Hence

$$\boxed{(\widehat{\mu}, \widehat{\sigma}^2) = (2, 5)}.$$

Question 6. Answer: C

The bias is

$$\text{Bias}(\widehat{\theta}_n) = \frac{1}{n}.$$

Therefore,

$$\text{MSE}(\widehat{\theta}_n) = \text{Var}(\widehat{\theta}_n) + \text{Bias}(\widehat{\theta}_n)^2 = \boxed{\frac{4}{n} + \frac{1}{n^2}}.$$

Question 7. Answer: B

Since

$$\text{SE}(\widehat{\theta}) \propto \frac{1}{\sqrt{n}},$$

the ratio of the new to old standard errors is

$$\frac{1/\sqrt{144}}{1/\sqrt{36}} = \frac{6}{12} = \frac{1}{2}.$$

Therefore the standard error is $\boxed{\text{divided by 2}}$.

Question 8. Answer: C

The probability that x_1 is not selected on one draw is $4/5$. Therefore the probability that it is absent from all five draws is

$$\left(\frac{4}{5}\right)^5.$$

Hence

$$P(x_1 \text{ appears at least once}) = \boxed{1 - \left(\frac{4}{5}\right)^5}.$$

Question 9. Answer: A

Increasing B increases the number of Monte Carlo draws from the empirical distribution. It therefore reduces simulation error in approximating the bootstrap distribution. It does not change the original dataset and cannot repair sampling bias or add population information.

Question 10. Answer: B

At the OLS optimum,

$$\boldsymbol{X}^\top \left(\boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}}\right) = \mathbf{0}.$$

Since

$$\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\widehat{\boldsymbol{\beta}},$$

the normal equations imply

$$\boxed{\boldsymbol{X}^\top \boldsymbol{e} = \mathbf{0}}.$$

Question 11. Answer: C

Up to a common positive factor, Cook's distance is proportional to

$$e_i^2 \frac{h_{ii}}{(1-h_{ii})^2}.$$

The four scores are

$$C_1 \propto 4\frac{0.10}{0.90^2} \approx 0.494,$$

$$C_2 \propto 1\frac{0.40}{0.60^2} \approx 1.111,$$

$$C_3 \propto 0.25\frac{0.70}{0.30^2} \approx 1.944,$$

and

$$C_4 \propto 9\frac{0.05}{0.95^2} \approx 0.499.$$

The largest score is for

$$\boxed{\text{Observation 3}}.$$

Question 12. Answer: A

A confidence interval for the conditional mean reflects uncertainty in estimating the mean response. A prediction interval for a future observation must additionally include the new observation's irreducible noise. Therefore the prediction interval is generally wider.

Question 13. Answer: B

The original regression model is nested inside the enlarged model: the new coefficient can always be set equal to zero. Therefore the minimum training RSS cannot increase. Since

$$R^2 = 1 - \frac{RSS}{TSS},$$

with the same responses and hence the same TSS, training R^2 cannot decrease.

Question 14. Answer: C

The model

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

is nonlinear as a function of x, but the parameters $\beta_0, \ldots, \beta_3$ enter linearly. It is therefore ordinary linear regression on the transformed feature vector

$$[1, x, x^2, x^3]^\top.$$

Question 15. Answer: A

At $x = 4$,

$$(4-1)_+^3 = 3^3 = 27, \qquad (4-3)_+^3 = 1^3 = 1.$$

Hence

$$\boxed{\boldsymbol{\phi}(4) = [1, 4, 16, 64, 27, 1]^\top}.$$

Question 16. Answer: C

The validation MSE values are

$$6.5, \quad 3.4, \quad 2.6, \quad 3.8.$$

The smallest is 2.6, corresponding to

$$\boxed{\text{degree } 5}.$$

Solution to Question 17: Quadratic Regression

The data are

$$x = (-2, -1, 0, 1, 2), \qquad y = (4, 1, 1, 3, 6).$$

(a) For the quadratic model,

$$\boldsymbol{\phi}(x) = \begin{bmatrix} 1 \\ x \\ x^2 \end{bmatrix}.$$

Therefore,

$$\boxed{\boldsymbol{X} = \begin{bmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}}.$$

(b) Using

$$\sum_i x_i = 0, \qquad \sum_i x_i^2 = 10, \qquad \sum_i x_i^3 = 0, \qquad \sum_i x_i^4 = 34,$$

we obtain

$$\boxed{\boldsymbol{X}^\top \boldsymbol{X} = \begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix}}.$$

(c) The required sums are

$$\sum_i y_i = 15,$$

$$\sum_i x_i y_i = (-2)(4) + (-1)(1) + (0)(1) + (1)(3) + (2)(6) = 6,$$

and

$$\sum_i x_i^2 y_i = 4(4) + 1(1) + 0(1) + 1(3) + 4(6) = 44.$$

Thus

$$\boxed{\boldsymbol{X}^\top \boldsymbol{y} = \begin{bmatrix} 15 \\ 6 \\ 44 \end{bmatrix}}.$$

(d) The normal equations are

$$\begin{bmatrix} 5 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 34 \end{bmatrix} \begin{bmatrix} \widehat{\beta}_0 \\ \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 15 \\ 6 \\ 44 \end{bmatrix}.$$

The second equation immediately gives

$$10\widehat{\beta}_1 = 6, \qquad \widehat{\beta}_1 = \frac{3}{5}.$$

The remaining equations are

$$5\widehat{\beta}_0 + 10\widehat{\beta}_2 = 15$$

and

$$10\widehat{\beta}_0 + 34\widehat{\beta}_2 = 44.$$

Solving gives

$$\widehat{\beta}_2 = 1, \qquad \widehat{\beta}_0 = 1.$$

Hence

$$\boxed{\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 1 \\ 3/5 \\ 1 \end{bmatrix}}$$

and the fitted function is

$$\boxed{\widehat{f}(x) = 1 + \frac{3}{5}x + x^2}.$$

(e) The fitted values and residuals are

x_i	y_i	$\widehat{y}_i$	$e_i = y_i - \widehat{y}_i$
-2	4	$19/5$	$1/5$
-1	1	$7/5$	$-2/5$
0	1	1	0
1	3	$13/5$	$2/5$
2	6	$31/5$	$-1/5$

Thus

$$\boxed{\widehat{\boldsymbol{y}} = \left(\frac{19}{5}, \frac{7}{5}, 1, \frac{13}{5}, \frac{31}{5}\right)^\top}$$

and

$$\boxed{\boldsymbol{e} = \left(\frac{1}{5}, -\frac{2}{5}, 0, \frac{2}{5}, -\frac{1}{5}\right)^\top}.$$

(f) The residual sum of squares is

$$RSS = \left(\frac{1}{5}\right)^2 + \left(-\frac{2}{5}\right)^2 + 0^2 + \left(\frac{2}{5}\right)^2 + \left(-\frac{1}{5}\right)^2 = \frac{10}{25} = \boxed{\frac{2}{5}}.$$

Therefore,

$$\boxed{\text{Training MSE} = \frac{RSS}{5} = \frac{2}{25} = 0.08}.$$

(g) At $x_0 = 1.5 = 3/2$,

$$\begin{aligned} \widehat{f}(1.5) &= 1 + \frac{3}{5}\left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)^2 \\ &= 1 + \frac{9}{10} + \frac{9}{4} \\ &= \boxed{\frac{83}{20}} = \boxed{4.15}. \end{aligned}$$

The regression function is nonlinear in the input x because it contains x^2. However, it is linear in the coefficient vector $(\beta_0, \beta_1, \beta_2)$, so after transforming the input to $[1, x, x^2]^\top$, the coefficients can be estimated by ordinary linear least squares.

Solution to Question 18: Bias–Variance Decomposition

For readability, write

$$f_0 = f(x_0), \qquad \widehat{f} = \widehat{f}_D(x_0), \qquad m = m(x_0) = \mathbb{E}_D[\widehat{f}].$$

(a) The bias of the fitted prediction at x_0 is

$$\boxed{\text{Bias}[\widehat{f}_D(x_0)] = \mathbb{E}_D[\widehat{f}_D(x_0)] - f(x_0) = m - f_0}.$$

(b) Since

$$Y_0 = f_0 + \varepsilon_0,$$

we have

$$Y_0 - \widehat{f} = f_0 + \varepsilon_0 - \widehat{f}.$$

Add and subtract m:

$$\boxed{Y_0 - \widehat{f} = \varepsilon_0 + (f_0 - m) + (m - \widehat{f})}.$$

The three terms represent future noise, systematic displacement of the average fitted model from the truth, and random variation of the fitted model around its own mean.

(c) Squaring,

$$\begin{aligned}(Y_0 - \widehat{f})^2 = \varepsilon_0^2 &+ (f_0 - m)^2 + (m - \widehat{f})^2 \\ &+ 2\varepsilon_0(f_0 - m) + 2\varepsilon_0(m - \widehat{f}) \\ &+ 2(f_0 - m)(m - \widehat{f}).\end{aligned}$$

Thus the three squared terms are

$$\varepsilon_0^2, \qquad (f_0 - m)^2, \qquad (m - \widehat{f})^2,$$

and the three cross terms are

$$2\varepsilon_0(f_0 - m),$$
$$2\varepsilon_0(m - \widehat{f}),$$

and

$$2(f_0 - m)(m - \widehat{f}).$$

(d) Take expectation over both the random training dataset D and the future noise ε_0.

First,

$$\mathbb{E}[\varepsilon_0] = 0.$$

Therefore,

$$\mathbb{E}\left[2\varepsilon_0(f_0 - m)\right] = 2(f_0 - m)\mathbb{E}[\varepsilon_0] = 0.$$

Because ε_0 is independent of D,

$$\begin{aligned}\mathbb{E}\left[2\varepsilon_0(m - \widehat{f})\right] &= 2\mathbb{E}[\varepsilon_0]\mathbb{E}_D[m - \widehat{f}] \\ &= 0.\end{aligned}$$

Finally,

$$\mathbb{E}_D[m - \widehat{f}] = m - \mathbb{E}_D[\widehat{f}] = m - m = 0,$$

so

$$\mathbb{E}\left[2(f_0 - m)(m - \widehat{f})\right] = 0.$$

Hence all three cross terms vanish.

(e) For the first squared term,

$$\mathbb{E}[\varepsilon_0^2] = \mathrm{Var}(\varepsilon_0) = \sigma^2.$$

The second squared term is

$$(f_0 - m)^2 = (m - f_0)^2 = \mathrm{Bias}[\widehat{f}_D(x_0)]^2.$$

For the third,

$$\begin{aligned}\mathbb{E}_D[(m - \widehat{f})^2] &= \mathbb{E}_D[(\widehat{f} - m)^2] \\ &= \mathrm{Var}_D[\widehat{f}_D(x_0)].\end{aligned}$$

Therefore,

$$\boxed{\mathbb{E}_{D,\varepsilon_0}\left[\left(Y_0 - \widehat{f}_D(x_0)\right)^2\right] = \sigma^2 + \mathrm{Bias}[\widehat{f}_D(x_0)]^2 + \mathrm{Var}_D[\widehat{f}_D(x_0)]}.$$

The three components are

$\boxed{\sigma^2}$ irreducible noise,

$\boxed{\mathrm{Bias}[\widehat{f}_D(x_0)]^2}$ squared bias,

and

$\boxed{\mathrm{Var}_D[\widehat{f}_D(x_0)]}$ variance.

(f) As model flexibility increases, squared bias often decreases because the model can represent a wider range of systematic patterns. At the same time, variance tends to increase because a more flexible fit is more sensitive to the particular training sample.

Initially, the reduction in squared bias may dominate, so test error decreases. After sufficient flexibility, the increase in variance may dominate, causing test error to rise. Training error can continue decreasing throughout because additional flexibility cannot worsen the best achievable fit to the same training data.